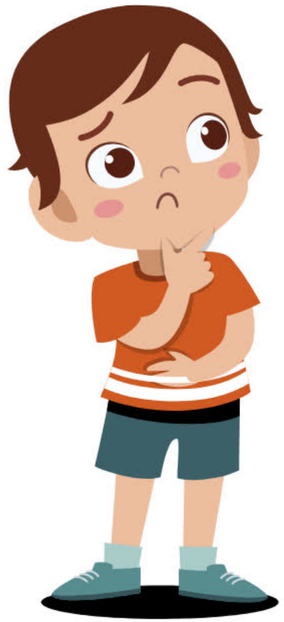
WHAT IS CLOUD?

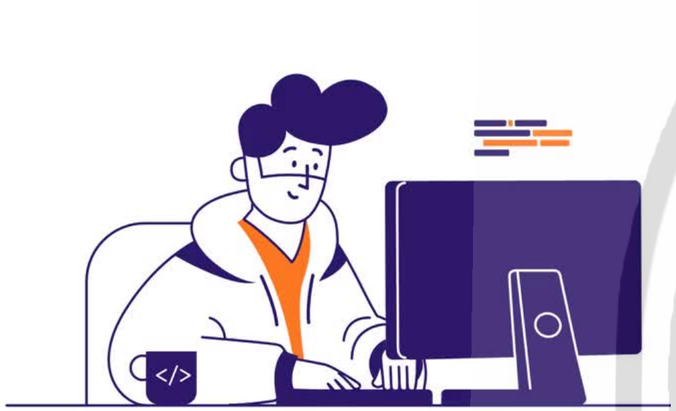


WHAT IS CLOUD?



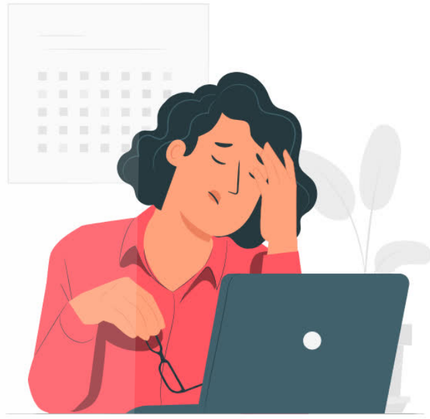
let's assume that you are watching a cloud in the sky. But have you ever thought that you can watch the same cloud anywhere in your location. This is how it works, you can access the data/application via the internet at any time in anyway

WHY CLOUD?









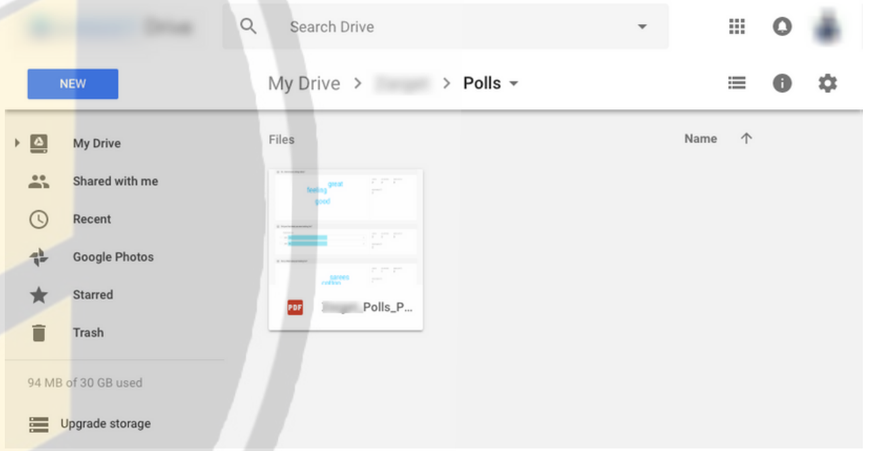
If you are working on a project for 30 days and you saved all your data on your local PC. But unexpectedly your PC got broken. You lost all your project data.

NOTE: Whenever you saved your data in a physical device like Memory cards, CDs, and pen drives. If you lost these devices or if the devices gets crashed then the data should not be retrieved

But what if you saved all your files in to cloud.







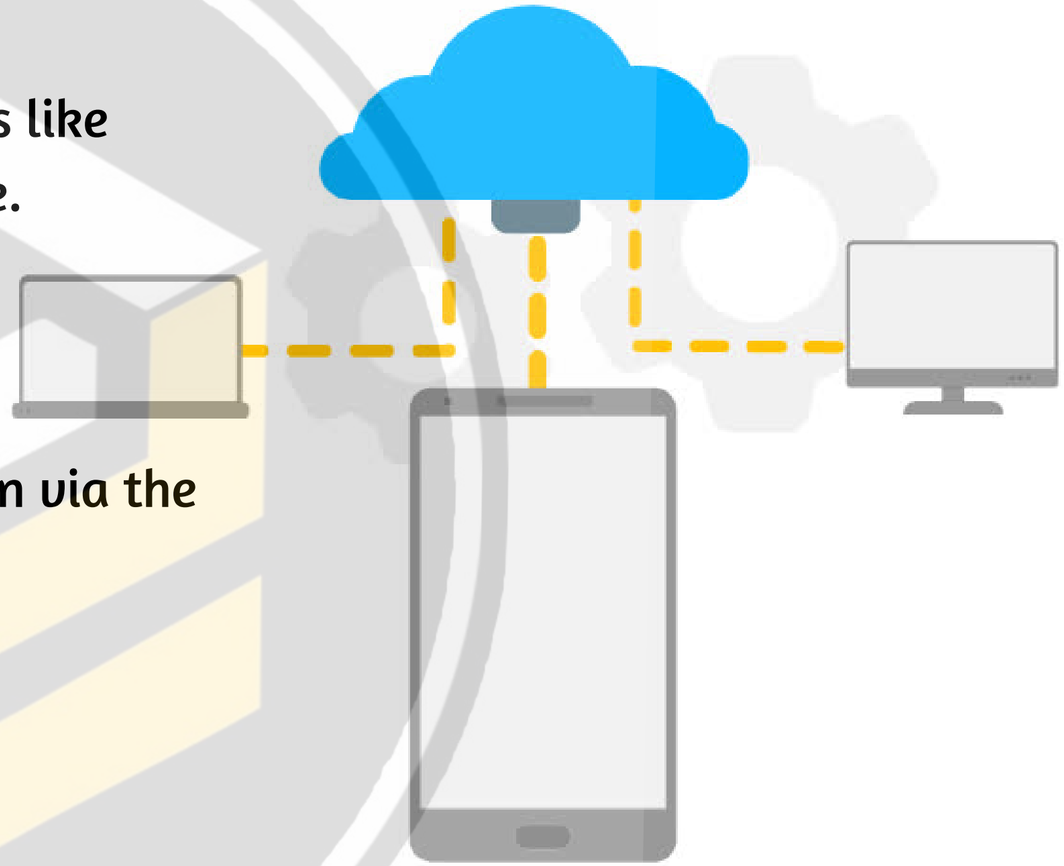
Now even if you lose your files now, no need to worry about data. Because all your files and folders are securely stored in the cloud.

WHAT IS CLOUD COMPUTING

Cloud Computing is the delivery of computing services like Servers, Storage, Database, Networking and software.

or

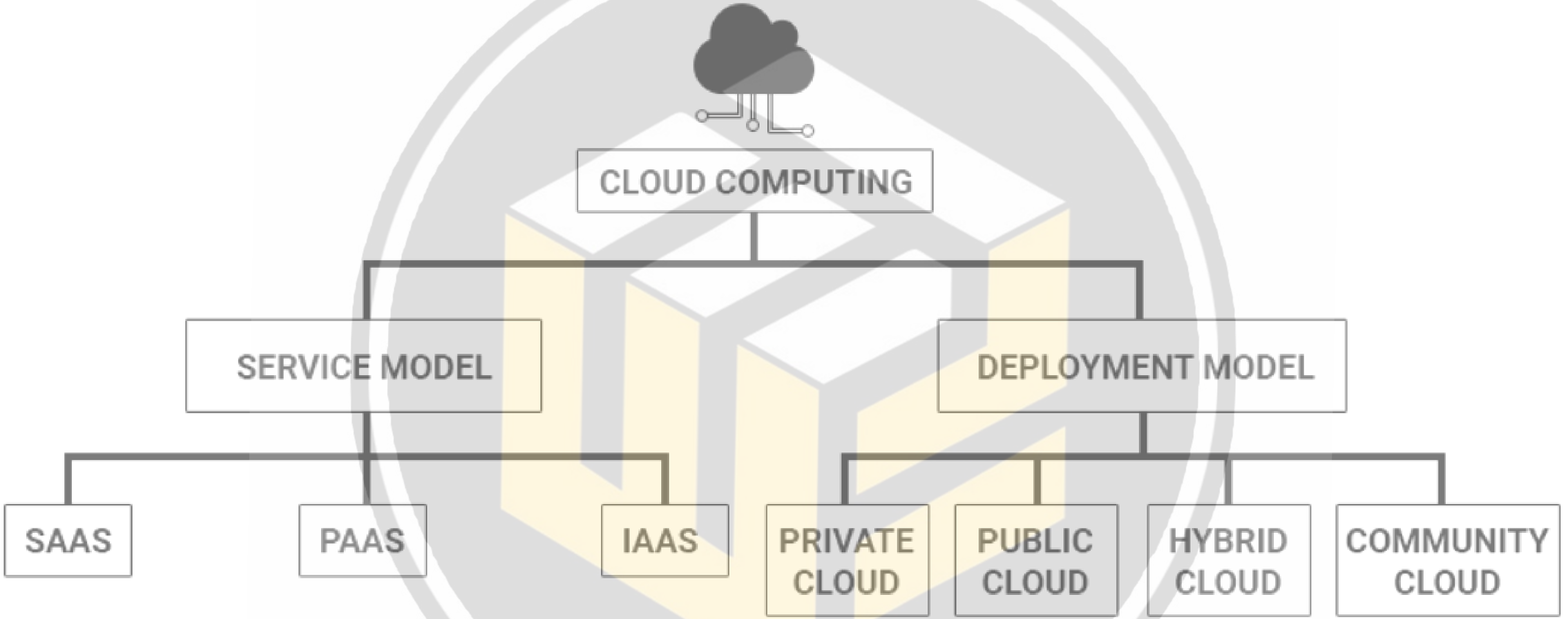
it is the processing of accessing the data or application via the internet



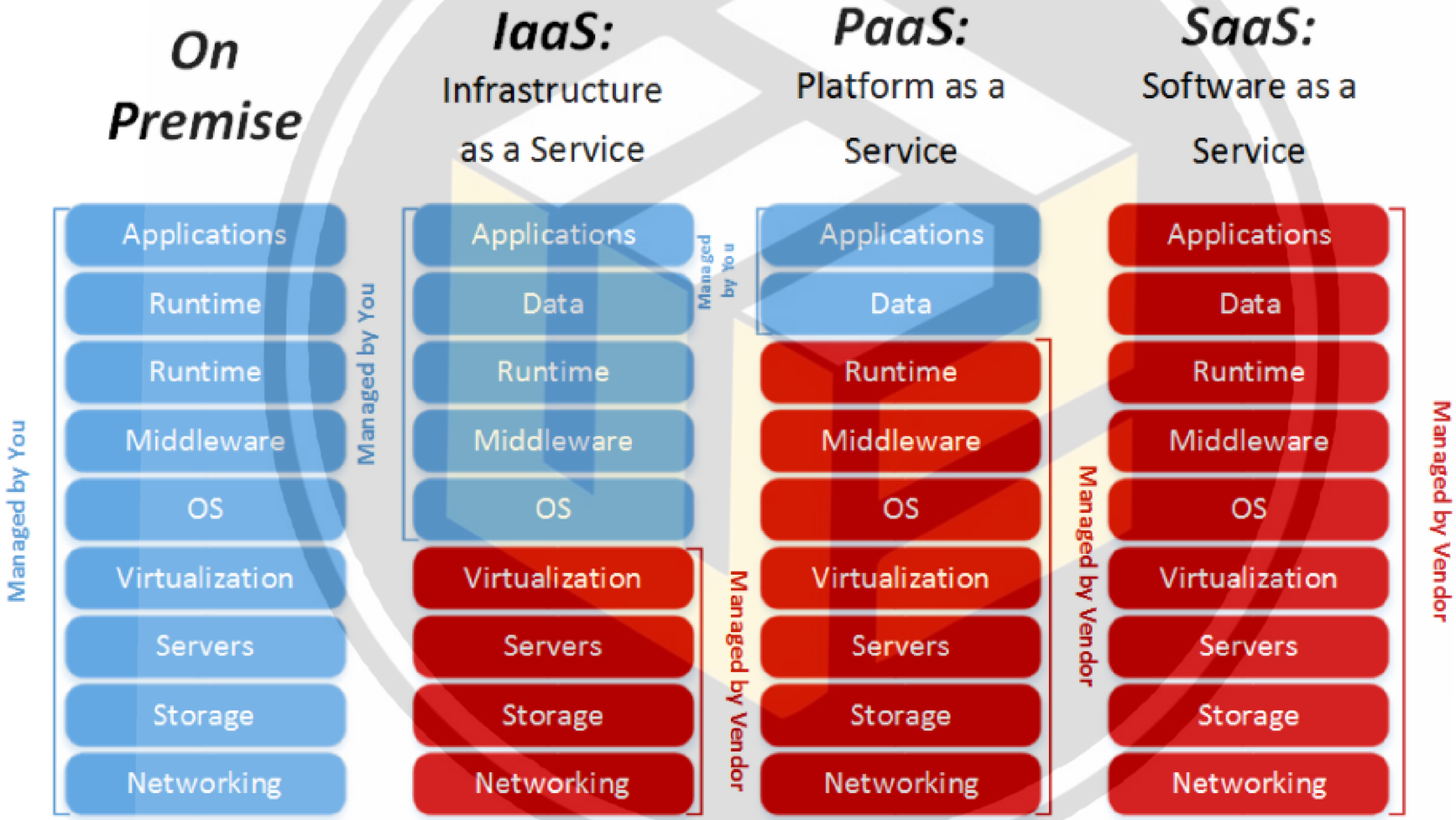
EX:

TYPES OF CLOUD COMPUTING




CLOUD COMPUTING
SERVICE MODEL
DEPLOYMENT MODEL
SAAS
PAAS
IAAS
PRIVATE CLOUD
PUBLIC CLOUD
HYBRID CLOUD
COMMUNITY CLOUD

SERVICE MODEL:



- **Infrastructure as a Service (IaaS):** In this model, the cloud provider offers virtualized computing resources, such as virtual machines, storage, and networking, to customers. Ex: **AWS, AZURE, GCP** - used for infrastructure provisioned and managed over internet



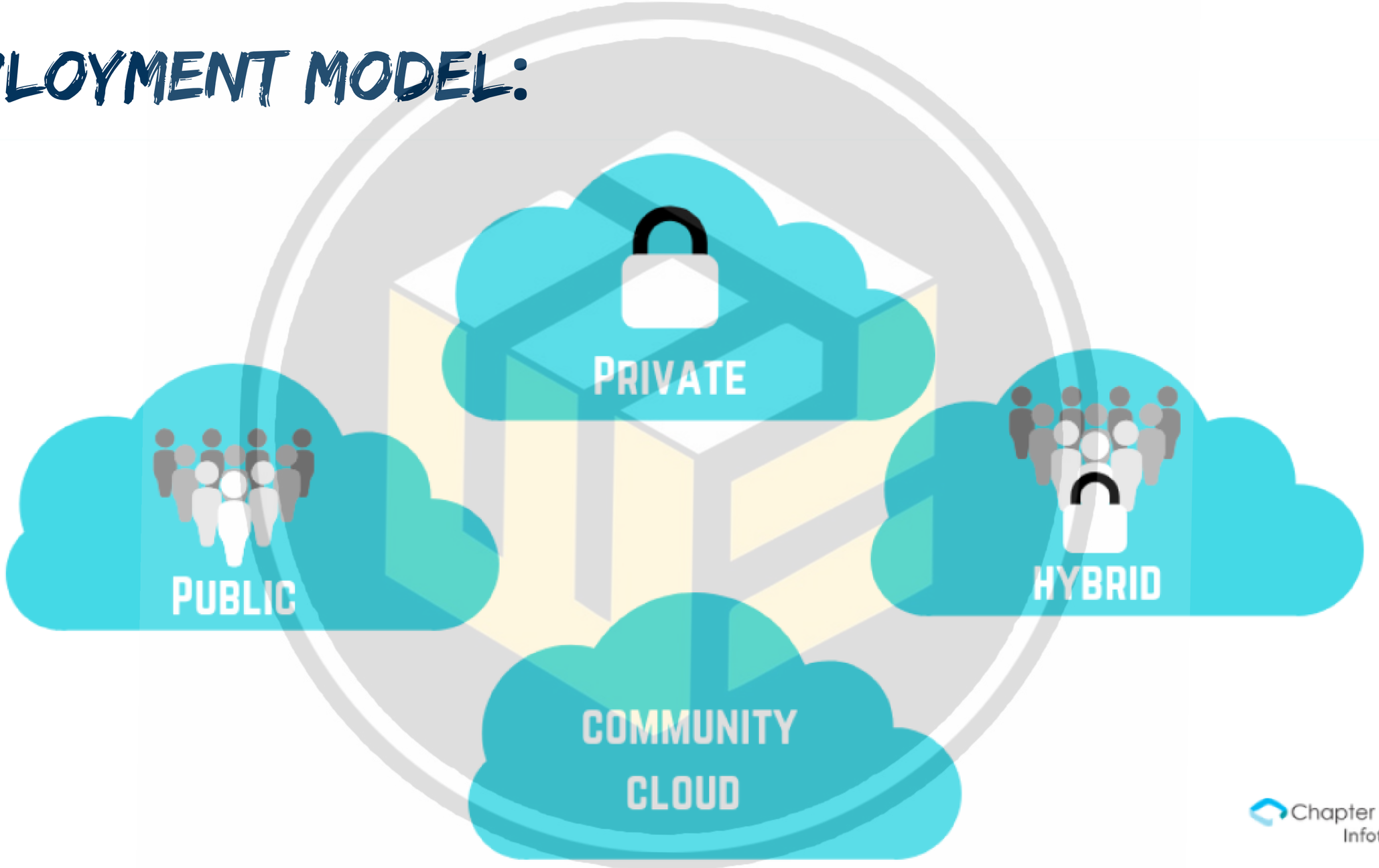




- **Platform as a Service (PaaS):** This model provides a platform for developers to develop, run, and manage their own applications, without having to worry about the underlying infrastructure.
  Ex: Heroku, Google App Engine, OCP, Godaddy and Microsoft Azure App Service

- **Software as a Service (SaaS):** SaaS is the most abstracted cloud computing type, delivering fully functional software applications via the internet. Users access these applications through a web browser. The provider hosts, maintains, and updates the software on behalf of the user.
  Ex: **Gmail -** You can manage inbox only, Google take cares of data centers, servers, network, storage maintenance etc .. All you need to worry about software and how you use it

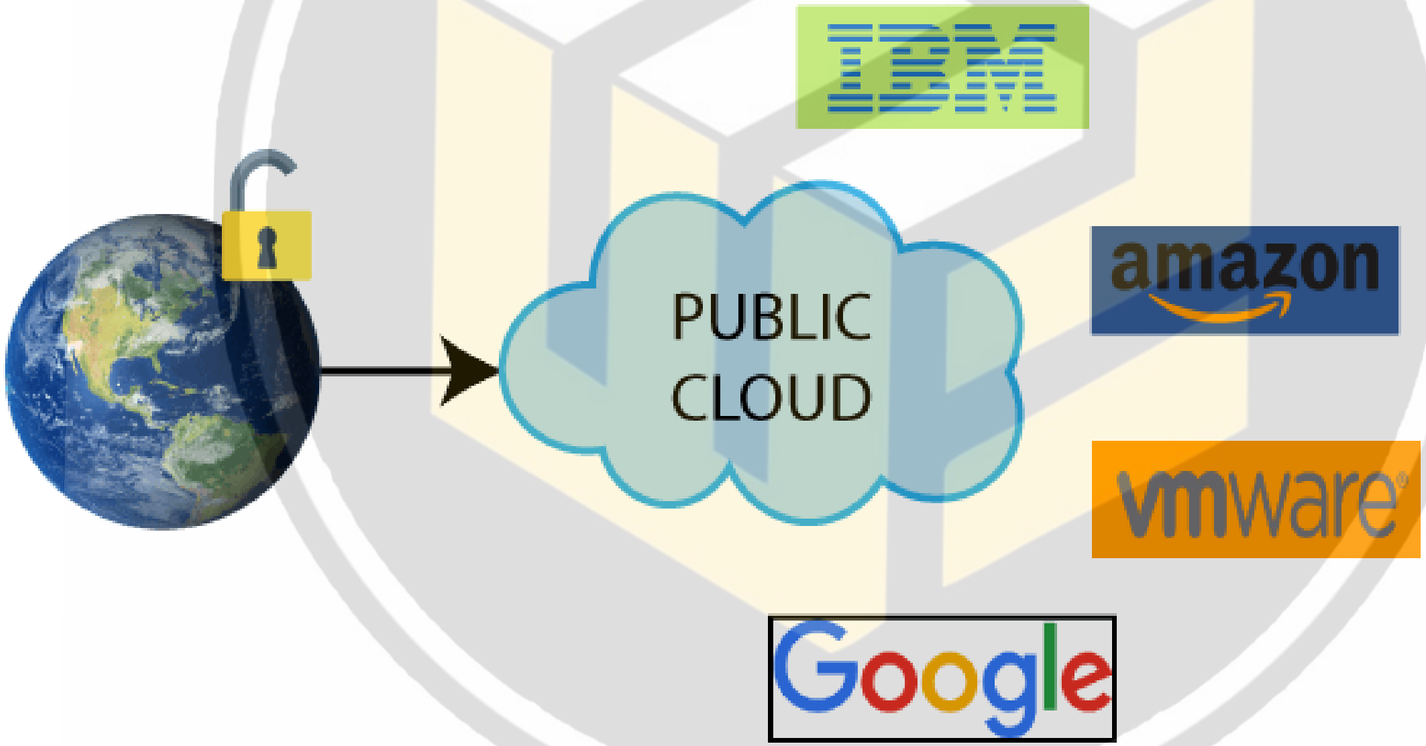
DEPLOYMENT MODEL:



Chapter 247
Infotech

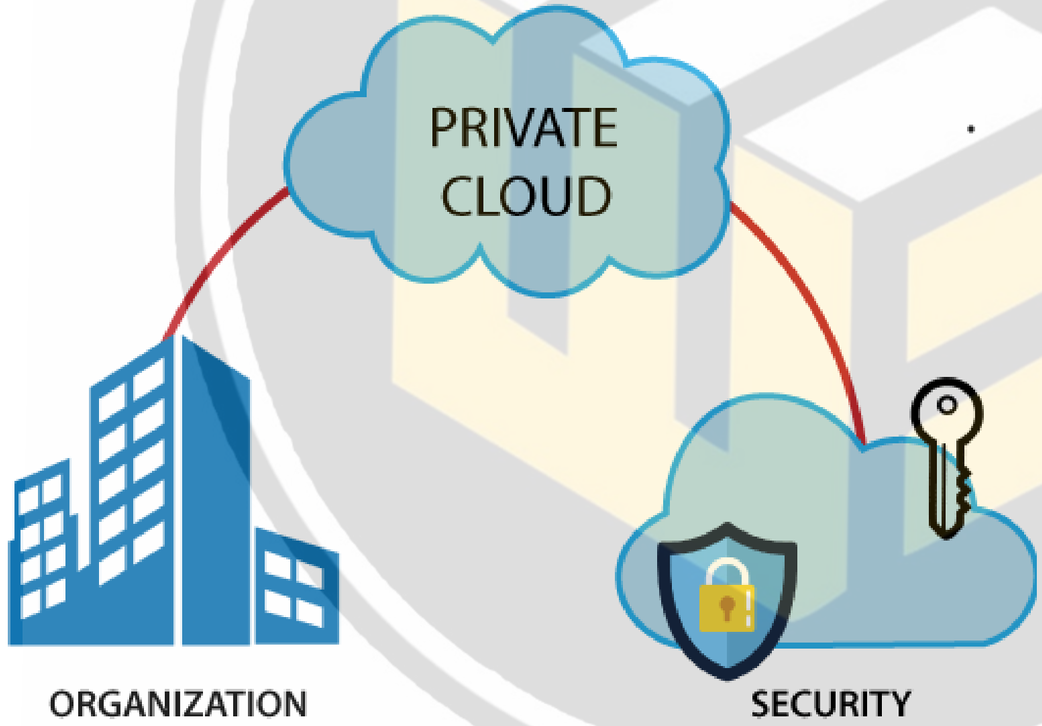
PUBLIC CLOUD:

Public cloud is open to all to store and access information via the Internet. Public clouds are managed by third parties. In public cloud, Security will be less when we compared to private and hybdrid.



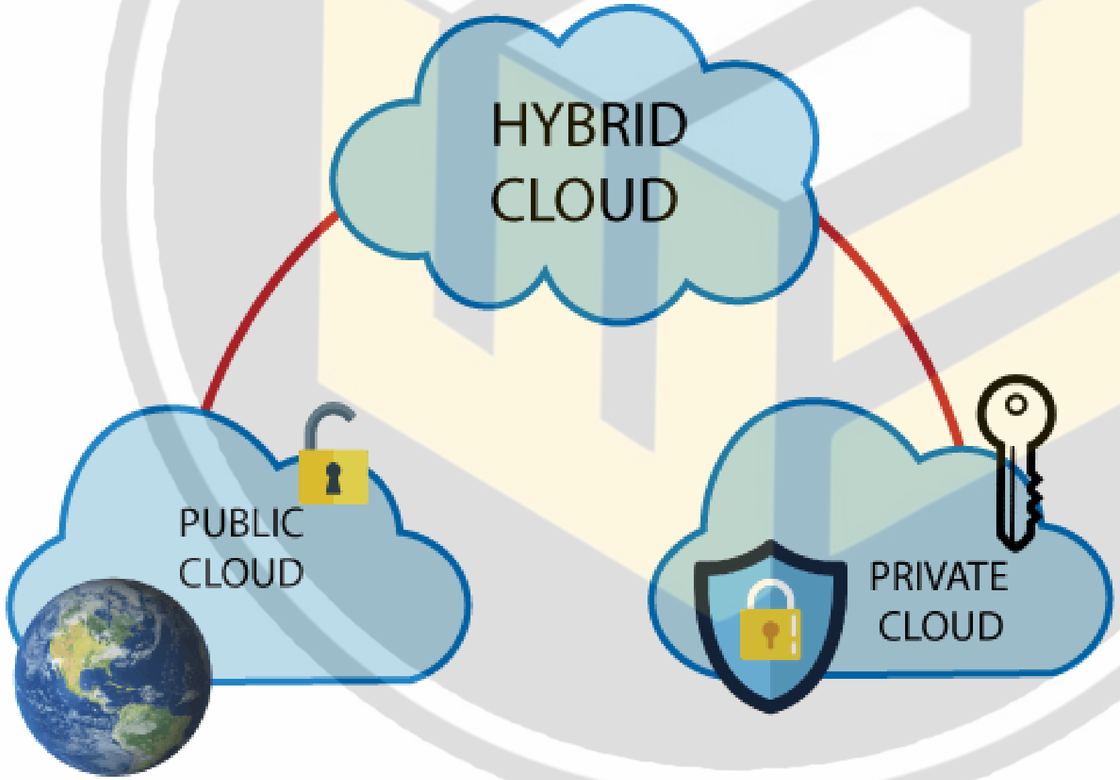
PRIVATE CLOUD:

A private cloud is also known as an internal cloud or corporate cloud. It is used by organizations to build and manage their data centers. In private cloud security will be high.



HYBRID CLOUD:

Hybrid cloud is the combination of both public cloud and private cloud.
If the services are running on public then it will not have much security.
If the services are running on private then the cloud have high security.



COMMUNITY CLOUD:

It allows multiple organizations to use same cloud to store their data.





TOP 10 CLOUD PROVIDERS















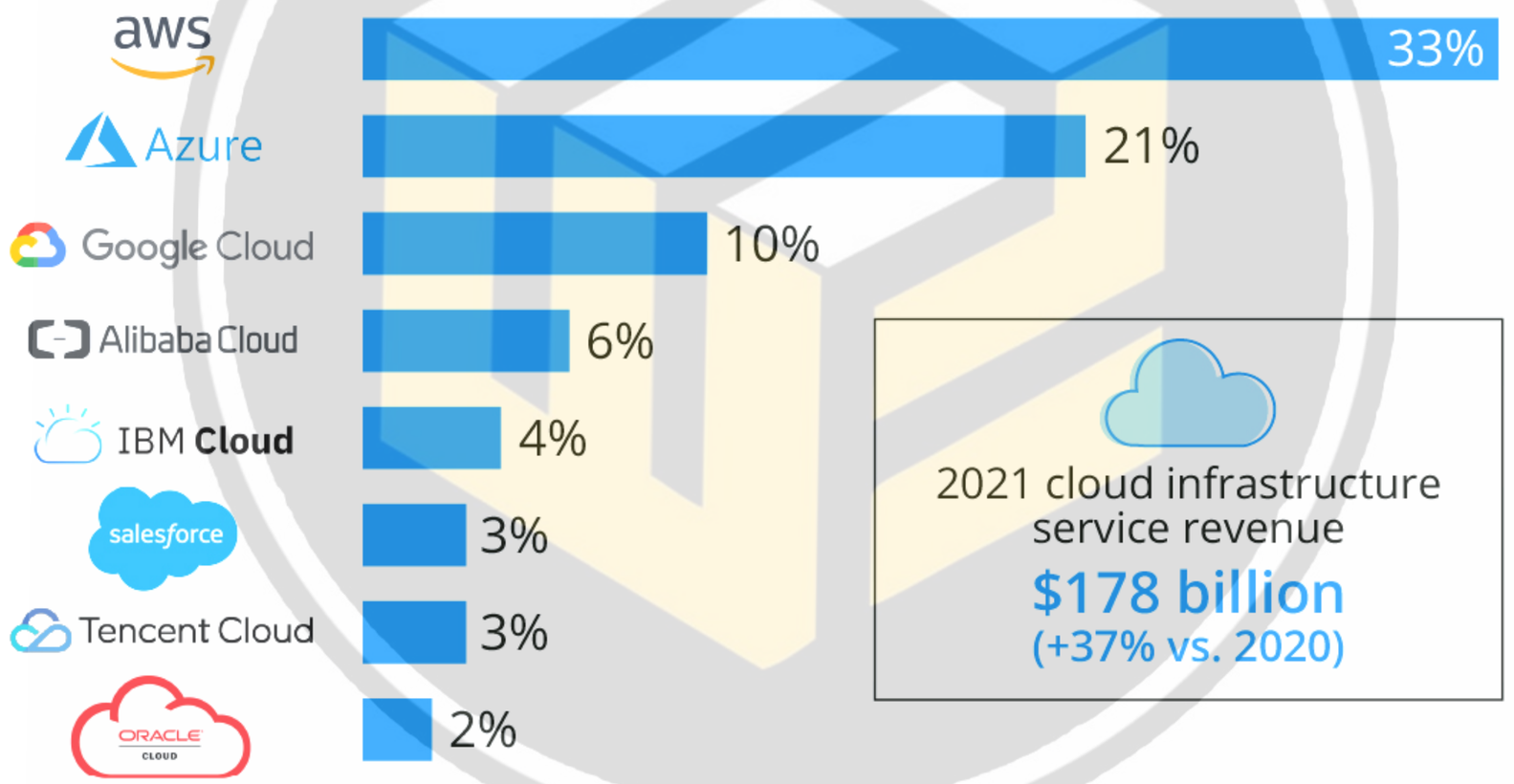






AMAZON LEADS $-180 BILLION CLOUD MARKET

WORLDWIDE MARKET SHARE OF LEADING CLOUD INFRASTRUCTURE SERVICE PROVIDERS IN Q4 2021*



WHAT IS



- AWS is abbreviated as Amazon Web Services.
- It is the best cloud provider
- It is the first among all the clouds
- It offers multiple services on different domains.
- It is the combination of SAAS, PAAS, IAAS.
- It is one of the most popular cloud computing platform. Now a days most of the companies are using this cloud because of numerous that allow them to store their data easily without the need for a physical space.

AWS WEB SERVICES LIST



WHY AWS IS SO POPULAR?

- AWS provides a wide range of cloud computing services that can be used to build and run applications.
- It can offer 200+ services on multiple domains.
- AWS covers about 31 geographic regions around the world.
- AWS provides a wide range of security features and compliance certifications that help customers to secure their applications and data in the cloud.
- The AWS Cloud spans 99 Availability Zones.
- It has 7 years of experience compared to another cloud.
- AWS allows users to select different operating systems, databases, and languages.
- It follows the Pay as you go, Model.

KEY TERMS:

"AVAILABILITY ZONE "



..aka "Data Center"

"REGION"



..aka "Geographical Area"

***Each Region consists of 2 or more availability zones*

"EDGE LOCATION"



...aka "AWS Endpoint used to cache content"

**Reduces latency to end user

PAY AS YOU GO MODEL



IF YOU GO TO BARBIQUE NATION RESTURANT



AND ATE ONLY 1 PLATE OF BIRYNI



SERVER ASKS YOU TO PAY 800RS

INSTEAD

IF YOU GO TO NORMAL RESTURANT THEY WILL CHARGE WHAT EVER YOU ORDER

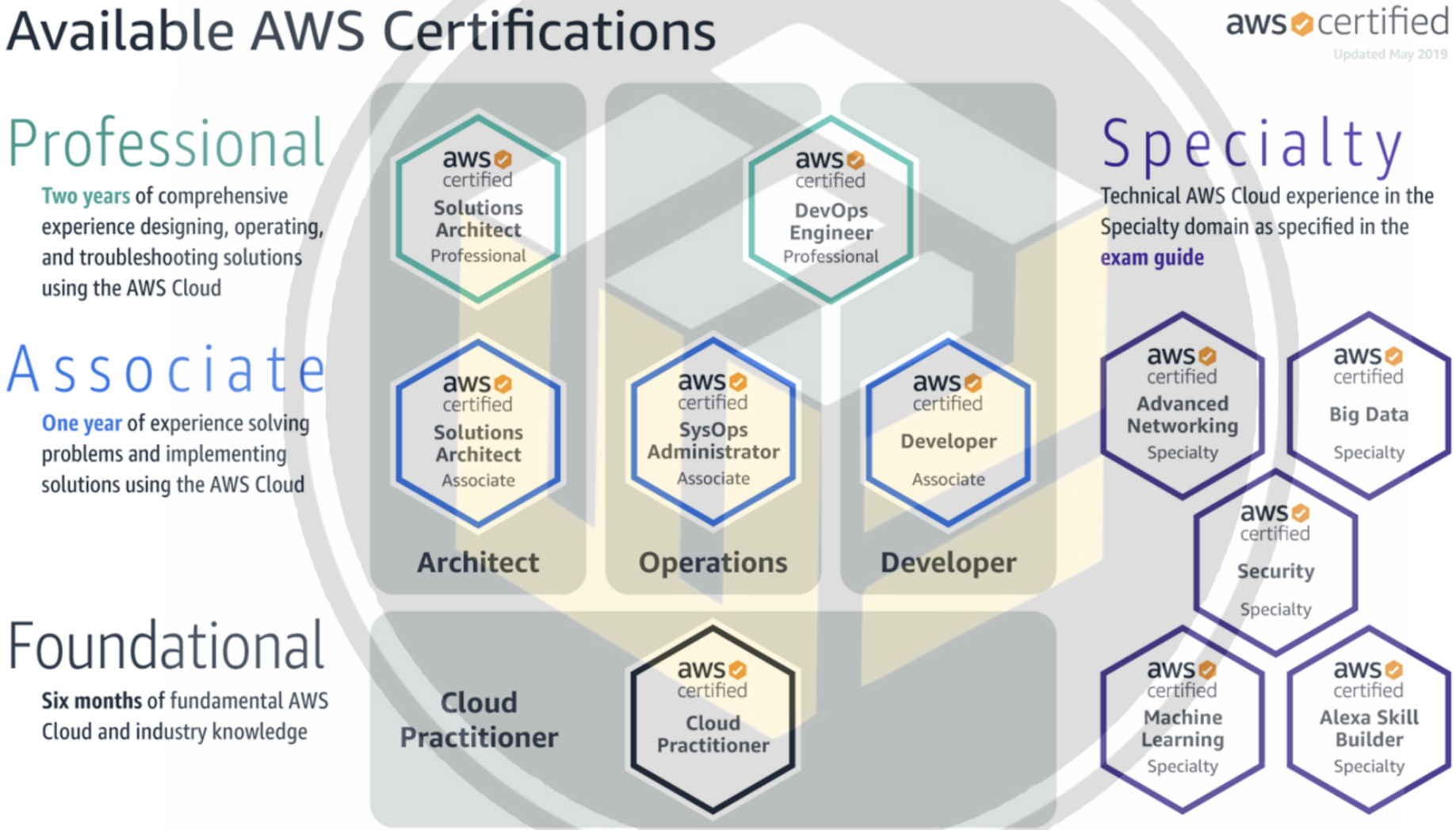
ITEM	PRICE
BIRYANI	$60.00
STARTERS	$65.00
MILK SHAKES	$80.00



ADVANTAGES OF AWS



AWS CERTIFICATIONS



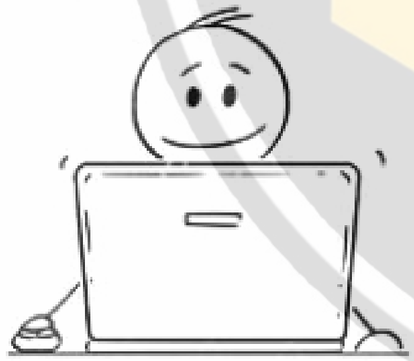

Thank You

WHAT IS DEVOPS

IN SOFTWARE DOMAIN, WE HAVE 2 TEAMS FOR DEVELOPING THE APPLICATION

PLAN
CODE
BUILD
TEST

DEVELOPMENT



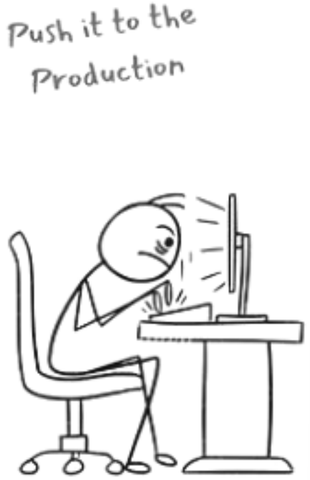
OPERATION

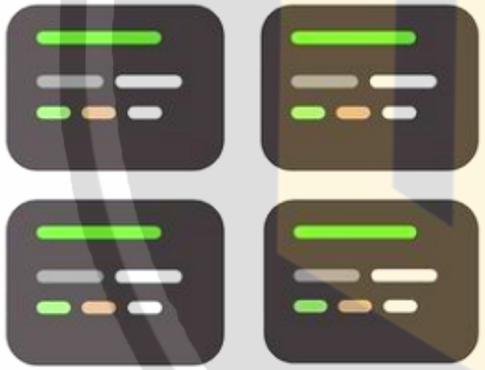


DEPLOY
OPERATION
MONITORING

DEVELOPMENT

THE DEV TEAM WRITES THE CODE FOR THE ENTIRE APPLICATION AND TOSSES IT TO OPERATIONAL TEAM FOR PRODUCTION





OPERATIONAL

THE OPS TEAM DEPLOYS, MONTOR OPERATE & MAINTAIN THE APPLICATION



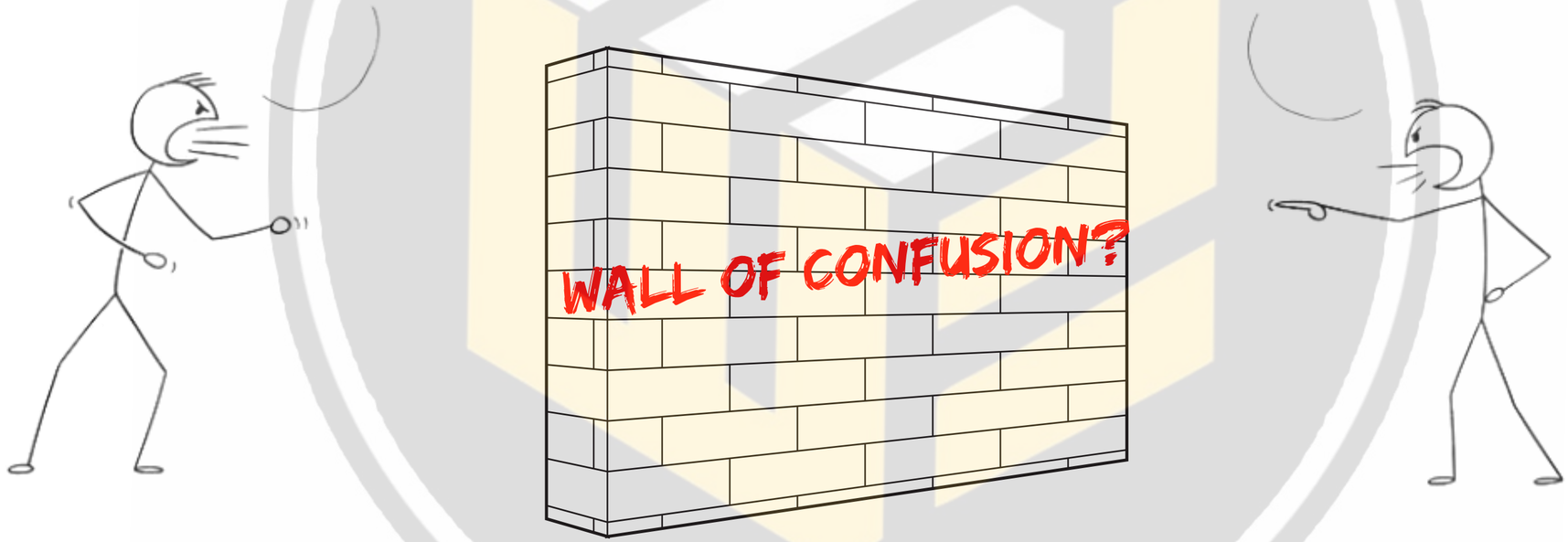


DEV TEAM

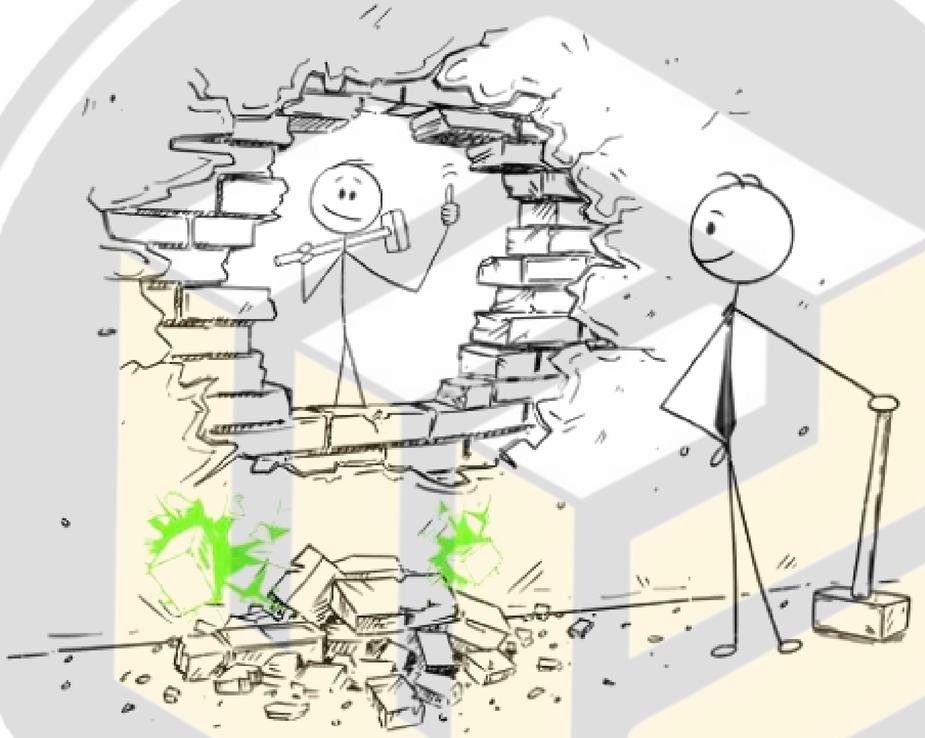
OPS TEAM

I Have given you the right code

I have deployed the code you given to me



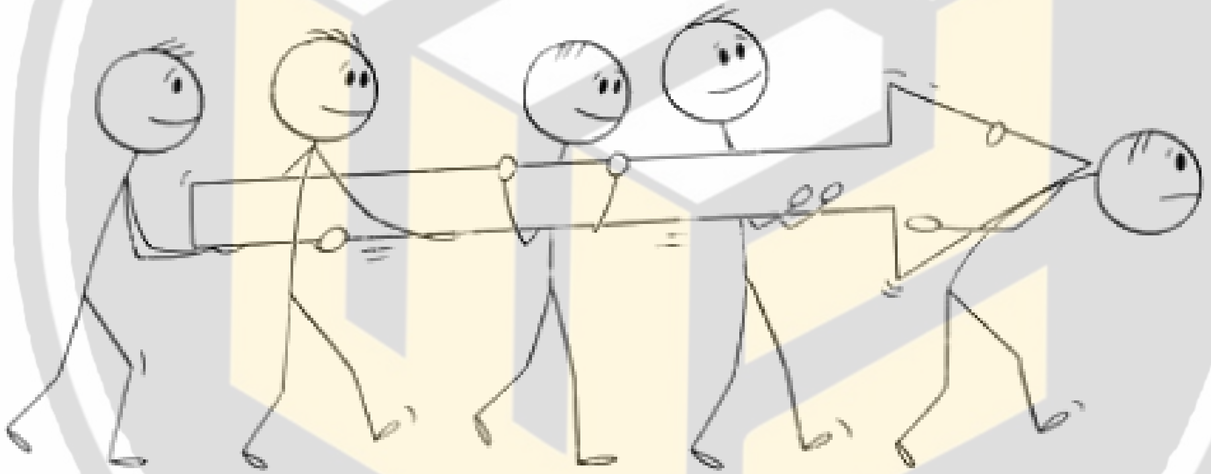
SO WHAT IF WE BREAK THE WALL?



THAT IS DEVOPS ---> DEV + OPS

DEVOPS IS NOT A TOOL, TECHNOLOGY OR FRAME WORK

DevOps is a cultural movement, mindset, philosophy to coordinate produce better, more reliable products



by automating infrastructure, workflow, and continuously measuring application performance for which they use a lot of tools

DEVOPS LIFE CYCLE

The Logo of DevOps represents infinite because its a never ending continuous process

